

Victoria Line Train Service Optimisation

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1. Introduction

The aim of this project is to determine the optimal train allocation across services on the Victoria Line of the London Underground to minimise passenger overcrowding, using TfL NUMBAT data. An integer linear programming model is developed to find the minimum number of trains that can be dispatched in each interval and direction whilst maintaining zero overcrowding.

The initial model however only minimised overcrowding. A sensitivity analysis, investigating the effect of a substantial reduction in train capacity revealed a solution where overcrowding was zero could still be found. Therefore the model was revised to minimise train dispatched instead and a sensitivity analysis was performed, investigating the effect of a decrease in train capacity on the minimum number of trains that can be dispatched so that overcrowding is zero.

The model assumes a fixed number of cars per train and average capacity per car. Trains on the Victoria Line consist of 8 cars with an average capacity per car of 107, so the average capacity per train is calculated to be 856. Additionally the victoria Line fleet size is 47, which is used as a constraint in the model.

2. Data

2.1 Data source

The model uses data from TfL NUMBAT 2025

<https://crowding.data.tfl.gov.uk/>.

The data on a typical autumn monday (NBT25MON) was chosen, simply due to it being the first day in the week. Link Loads data was used.

Information used in determining the model constraints was gathered from the following websites:

1. <https://tfl.gov.uk/corporate/transparency/freedom-of-information/foi-request-detail?referenceId=FOI-4157-2425>
2. <https://content.tfl.gov.uk/06-lu-rolling-stock.pdf>
3. <https://mycommute.london/victoria>

2.2 Data preparation

In the data preparation phase columns not relevant to the model were dropped and the Victoria Line was selected. Then the data frame was converted to a long format, time intervals were initially arranged in columns but the process converted them to rows. Finally the order link numbers were then adjusted - they were kept the same for NB but reversed for SB as instead of order the numbers were going to be allocated to links.

3. Mathematical Model

3.1 Model 1

Let $t = 1, \dots, T$ denote each 15-minute interval

Let $I = 1, \dots, 15$ where $i \in I$ denotes the link number

1. Brixton - Stockwell
2. Stockwell - Vauxhall
3. Vauxhall - Pimlico
4. Pimlico - Victoria
5. Victoria - Green Park
6. Green Park - Oxford Circus
7. Oxford Circus - Warren Street
8. Warren Street - Euston
9. Euston - King's Cross St Pancras
10. King's Cross St Pancras - Highbury & Islington
11. Highbury & Islington - Finsbury Park
12. Finsbury Park - Seven Sisters
13. Seven Sisters - Tottenham Hale
14. Tottenham Hale - Blackhorse Road
15. Blackhorse Road - Walthamstow Central

Decision variables

$O_{i,t}^d$ – overcrowding in direction $d \in \{\text{NB}, \text{SB}\}$ at link i during interval t

x_t^{NB} – the number of trains dispatched from Brixton during interval t

x_t^{SB} – the number of trains dispatched from Walthamstow Central during interval t

Parameters

$L_{i,t}^d$ – the number of passengers travelling on link i in direction $d \in \text{NB}, \text{SB}$

c – capacity per train

F – fleet size

Objective function

$$\text{Minimise } \sum_{d \in \text{NB}, \text{SB}} \sum_{i=1}^{15} \sum_t^T O_{i,t}^d$$

Train circulation

Trains are modeled following a cyclical structure. Each train is either dispatched at Brixton or Walthamstow Central and follows the route until the end of the line and then turns around to reach the other end of the line, thus completing a cycle. It is assumed it takes 30 minutes to travel the full line, so takes 60 minutes to complete a cycle.

The line is split into halves to be compatible with the 15-minute intervals used in the NUMBAT dataset.

It is also assumed at $t = 0$ there is one train in circulation in each half of the line in each direction.

$$x_t = x_t = 0 \text{ for } t \leq -2$$

$$x_t = x_t = 1 \text{ for } t \in -1, 0$$

Capacity constraints

Link load is the demand on trains between platforms. The model defines overcrowding as the excess passenger demand over the provided capacity in the relevant link and interval.

$$O_{i,t}^{NB} \geq L_{i,t}^{NB} - c(x_t^{NB} + x_{t-2}^{SB}), \forall (i \leq 8 \cap t > 2)$$

$$O_{i,t}^{NB} \geq L_{i,t}^{NB} - c(x_{t-1}^{NB} + x_{t-3}^{SB}), \forall (i \geq 9 \cap t > 3)$$

$$O_{i,t}^{SB} \geq L_{i,t}^{SB} - c(x_t^{SB} + x_{t-2}^{NB}), \forall (i \geq 9 \cap t > 2)$$

$$O_{i,t}^{SB} \geq L_{i,t}^{SB} - c(x_{t-1}^{SB} + x_{t-3}^{NB}), \forall (i \leq 8 \cap t > 3)$$

Fleet constraint

$$4 + \sum_{\tau=t-3}^t (x_{\tau}^{NB} + x_{\tau}^{SB}) \leq F, \forall t$$

Frequency constraint

Peak frequency (the most trains on the network - in one direction) is 36 trains per hour. Since the model used 15 minute intervals we have

$$36 \times \frac{15}{60} = 9$$

$$x_t^{NB} \leq 9 \text{ and } x_t^{SB} \leq 9$$

Integer constrain

$$x_t^d \in \mathbb{Z}_{\geq 0}$$

3.2 Model 2

The second model is very similar, following the same constraints with the addition of overcrowding being set to 0.

Decision variables

x_t^{NB} – the number of trains dispatched from Brixton during interval t

x_t^{SB} – the number of trains dispatched from Walthamstow Central during interval t

Parameters

$L_{i,t}^d$ – the number of passengers travelling on link i in direction $d \in \{NB, SB\}$

c – capacity per train

F – fleet size

Objective function

$$\text{Minimise } \sum_{d \in \{NB, SB\}} \sum_{t=1}^T x_t^d$$

Constrains

$$\begin{aligned}
x_t &= x_t = 0 \text{ for } t \leq -2 \\
x_t &= x_t = 1 \text{ for } t \in -1, 0 \\
L_{i,t}^{NB} &\leq c(x_t^{NB} + x_{t-2}^{SB}), \forall (i \leq 8 \cap t > 2) \\
L_{i,t}^{NB} &\leq c(x_{t-1}^{NB} + x_{t-3}^{SB}), \forall (i \geq 9 \cap t > 3) \\
L_{i,t}^{SB} &\leq c(x_t^{SB} + x_{t-2}^{NB}), \forall (i \geq 9 \cap t > 2) \\
L_{i,t}^{SB} &\leq c(x_{t-1}^{SB} + x_{t-3}^{NB}), \forall (i \geq 8 \cap t > 3) \\
4 + \sum_{\tau=t-3}^t (x_\tau^{NB} + x_\tau^{SB}) &\leq F, \forall t \\
x_t^{NB} &\leq 9 \\
x_t^{SB} &\leq 9 \\
x_t^d &\in \mathbb{Z}_{\geq 0}
\end{aligned}$$

4. Optimisation Approach

The model was implented in Python using Pyomo and solved using GPLK at first and then HiGHS when a more powerful solver was needed. The initial model was implemented and a solution where overcrowding equaled zero was found for the intial capacity of 856. However after adjusting the capacity down to even 428, the model still found a solution where overcrowding was equal to zero. Therefore the model was instead adjusted to minimise the trains dispatches needed for overcrowding to remain at zero.

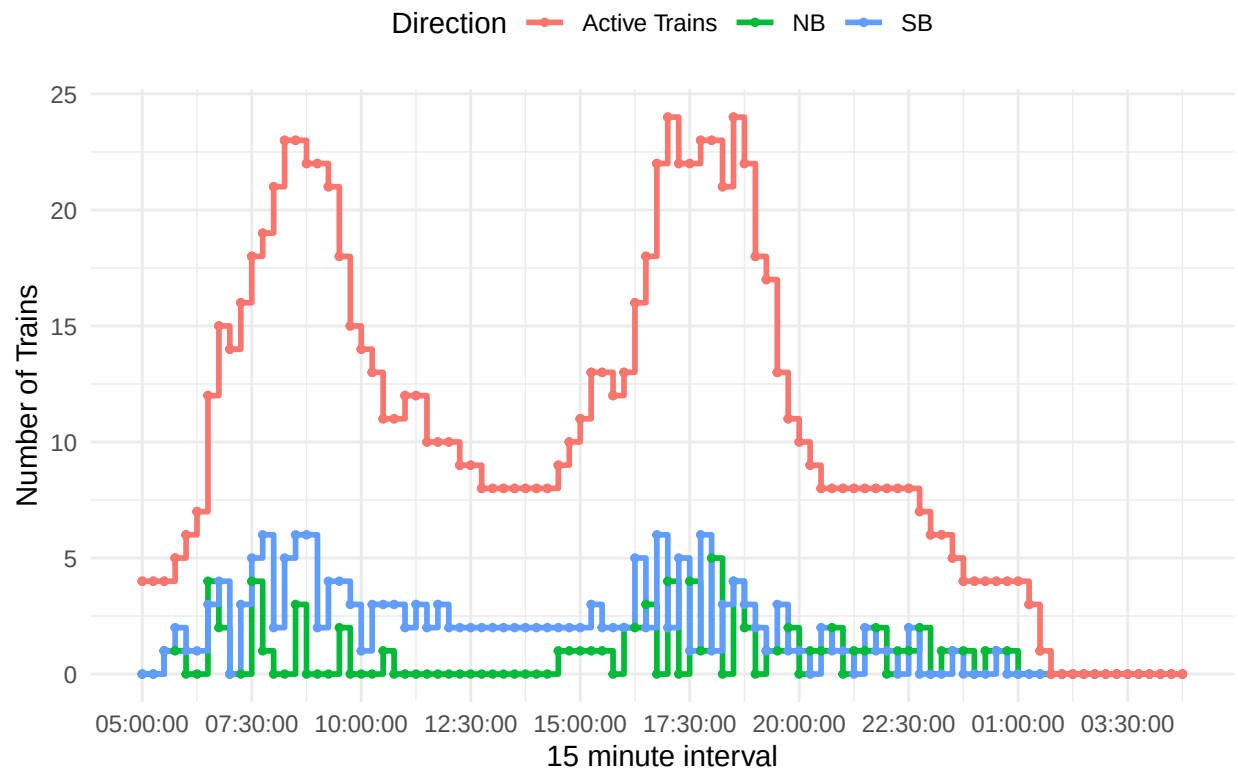
5. Results

5.1 Optimal dispatch schedule and Active fleet

Table 1: Snippet of dataframe produced

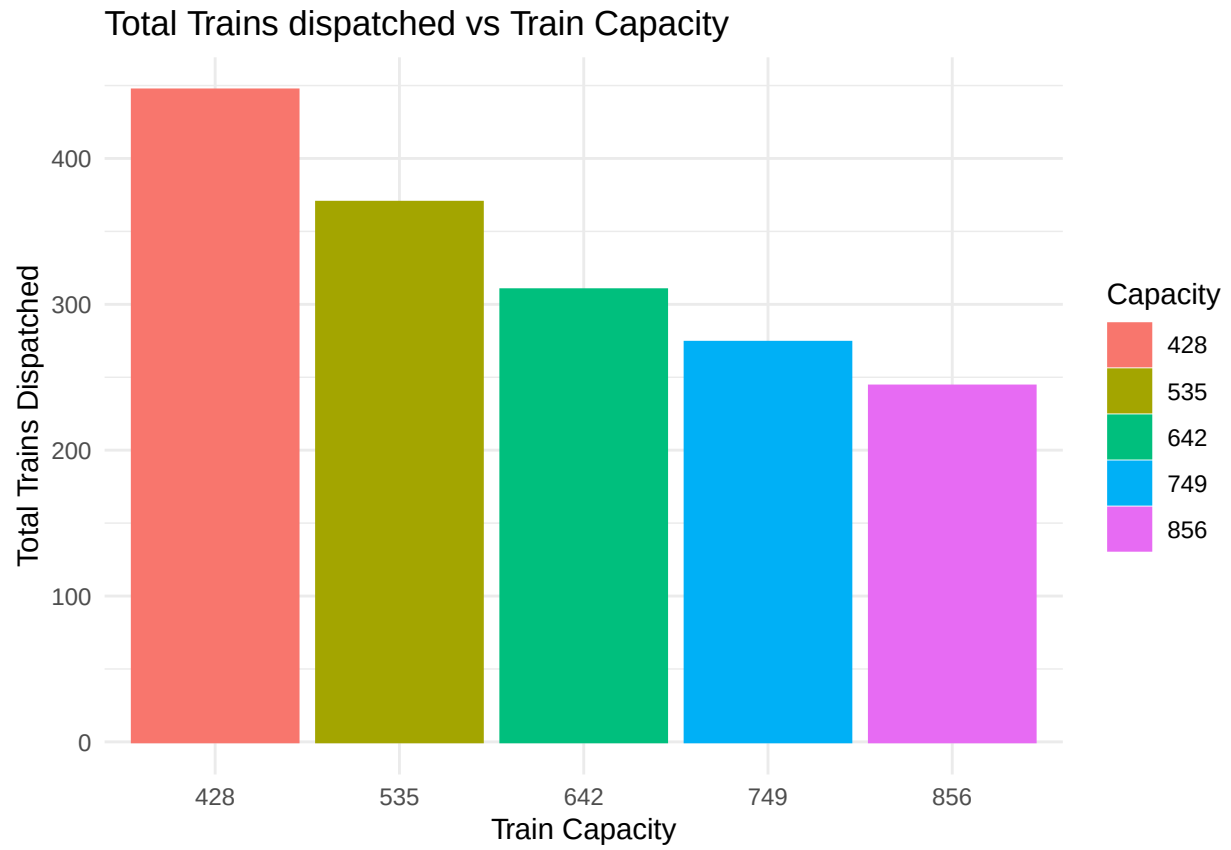
Time	t	NB	SB	Trains_active
0500-0515	1	0	0	4
0515-0530	2	0	0	4
0530-0545	3	1	1	4
0545-0600	4	1	2	5
0600-0615	5	0	1	6
0615-0630	6	0	1	7
0630-0645	7	4	3	12
0645-0700	8	2	4	15
0700-0715	9	0	0	14
0715-0730	10	0	3	16
0730-0745	11	4	5	18
0745-0800	12	1	6	19
0800-0815	13	0	2	21
0815-0830	14	0	5	23
0830-0845	15	3	6	23

Optimised Train Dispatches



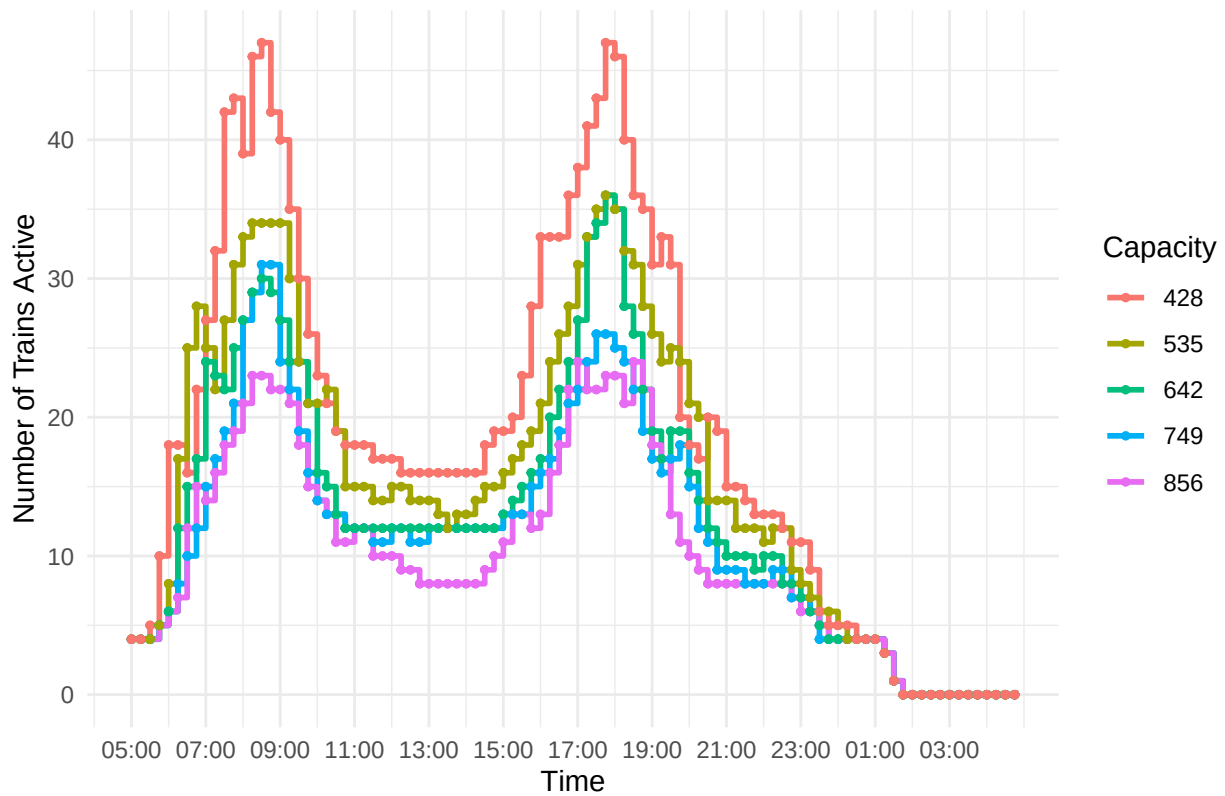
6. Sensitivity Analysis

6.1 Effect of train capacity

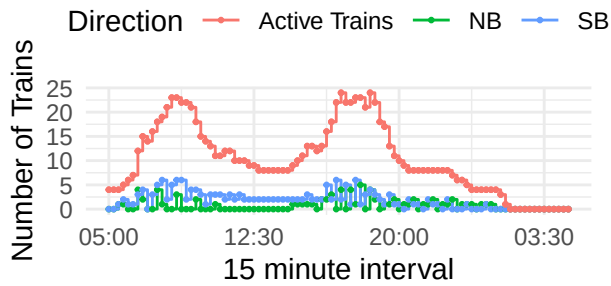


6.2 Effect on fleet utilisation

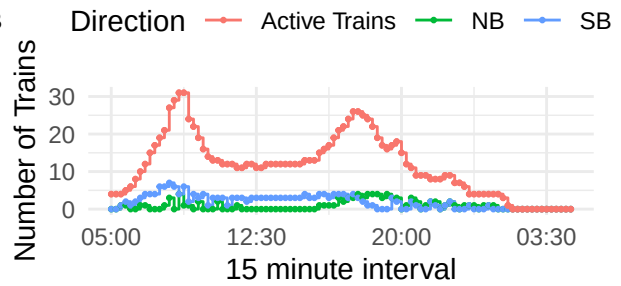
Minimised Active Trains for each Capacity and Interval



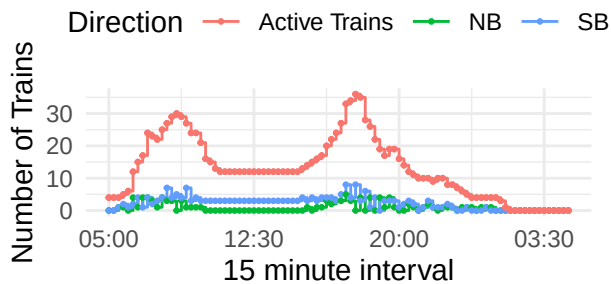
Optimised Train Dispatches at capacity 856



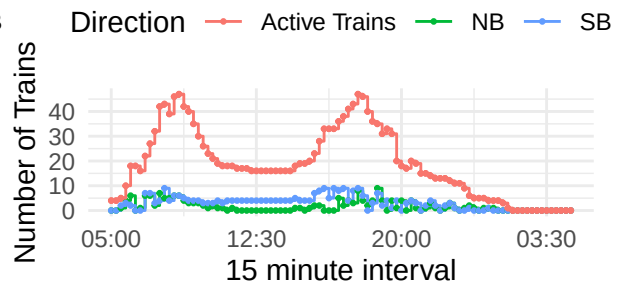
Optimised Train Dispatches at capacity 749



Optimised Train Dispatches at capacity 642



Optimised Train Dispatches at capacity 428



7. Limitations
8. Conclusion